Section C: Development Environment and Resources

Project 06 - Game of Life with Music | Prepared by: Ethan Jarren Diego

1. Python version, IDE and install steps

1. Install **Python 3.10 or newer** from python.org. Tick **Add Python to PATH** during setup.

2. Install an IDE: **VS Code** (with the Python extension) or **PyCharm**.

3. Open a terminal in the project folder and run: `pip install pygame numpy`

4. Check the versions: `python --version` and `pip show pygame numpy`. Take a screenshot of the result.

5. Accept Vik's GitHub invite, clone the repo, add `sound.py`, then commit and push.

2. Libraries table

Library	What it does in our app
pygame	Opens the window, draws the grid, reads mouse and keyboard input, and plays the audio.
numpy	Builds the musical tones (sound waves) as arrays of numbers.
unittest	Built in to Python. Runs our tests so we know the rules work.

3. Formulas and resources

• **B3/S23 rule:** a dead cell with exactly 3 living neighbours is **born** (B3). A living cell with 2 or 3 living neighbours **survives** (S23). Every other cell dies or stays dead.

• **Note frequency:** `f = 440 x 2^((m - 69) / 12)`, where m is the MIDI note number (m = 69 is A4 = 440 Hz). Every 12 steps doubles the frequency (one octave).

• **Pattern references: Block** (still life, never changes), **Blinker** (flips between a row and a column every generation), **Glider** (moves diagonally across the grid).

• **Dataset:** none needed.

4. App organization (folders and files)

Rule: **one file per job**, and the computation is kept separate from the interface (like a kitchen and a dining room: the cooking does not happen at the tables).

```
game_of_life/
  main.py            - starts the app
  gui.py             - pygame window, grid drawing, buttons, sliders, status bar
  life.py            - Game of Life rules (B3/S23), no pygame here
  patterns.py        - block, blinker, glider
  sound.py           - numpy tones and audio playback
  tests/
    test_life.py     - unittest for the rules
  docs/pr1/          - PDF and wireframe for Progress Report 1
```

Because life.py never imports pygame, we can test the rules without opening a window.